

7B-Scale Validation

Beating DISP-LLM, training discipline at scale, and what downstream eval actually shows

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- Everything here is the **same pruner** (row-encoder + BiLSTM, $\approx 2\text{M}$ params) that ran on GPT-2/OPT-125M — no architecture change, only the weight-extraction plumbing differs per model.
- Three real 7B-scale sweeps: **Llama-2-7B / WikiText-2**, **Mistral-7B / C4**, **Mistral-7B / WikiText-2**. Plus one first look at a downstream-task objective (**CNN/DailyMail**).

Headline

On the one directly matched comparison available in the literature (Llama-2-7B, WikiText-2, no-weight-update), we **beat DISP-LLM's own published numbers at every converged operating point**. Downstream zero-shot accuracy tells a more complicated story — covered honestly, not smoothed over.

Training regime & the convergence check

The objective, unchanged from GPT-2/OPT-125M

Frozen base model, only the pruner ϕ trains:

$$\mathcal{L} = (\text{CE}_{\text{pruned}} - \text{CE}_{\text{orig}}) + \lambda \cdot \overline{\text{gate}}$$

- Forward pass runs **twice** per step — once ungated (CE_{orig} , no_grad), once gated ($\text{CE}_{\text{pruned}}$, gradients flow only into ϕ).
- λ is the sole sparsity/accuracy knob, swept per model.
- Everything at 7B scale trains this way with *no per-model tuning* beyond the weight-extraction plumbing.

► What's new at this scale isn't the objective — it's needing to know *when to stop*, cheaply, without a hand-picked step count.

That's the convergence check.

Convergence check: block-mean flatness, not a fixed step count

Every `check_every` steps, compare the last window
block-means of each layer's keep-rate history against
the most recent one:

$$\bar{k}_\ell(cp) = \frac{1}{\text{check_every}} \sum_{s=cp-\text{check_every}}^{cp} k_\ell(s)$$

$$\text{tol} = \max(\text{rel_tol} \cdot |\bar{k}_\ell^{\text{ref}}|, \text{abs_tol})$$

Raw signal fires only if *every layer* is within `tol` of
the reference, for *every* checkpoint in the window,
after a `burn_in` floor.

Why per-layer, not global

A global average can look flat while
individual layers are still trading pruned
neurons back and forth — the check has
to fail if *any* layer is still moving.

B9: don't trust the first flat window

A raw signal can be a plateau, or just noise sitting still under a constant LR. So it doesn't confirm convergence — it triggers a **cosine LR decay** first:

$$\text{lr}(s) = \text{lr}_{\min} + \frac{1}{2}(\text{lr}_0 - \text{lr}_{\min}) \left(1 + \cos \frac{\pi s}{W}\right)$$

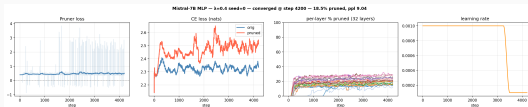
After W steps at $\text{lr}_{\min}$, **re-check**. Passes $\Rightarrow$ confirmed converged. Fails $\Rightarrow$ hold at $\text{lr}_{\min}$, keep training (post_decay) — the original signal was noise the higher LR was masking.

pre_decay $\rightarrow$ decaying $\rightarrow$ {CONFIRMED,
post_decay}

► This three-state machine is what lets every sweep in this talk report “converged” as a checked fact, not an assumption.

**Mistral-7B: does the method just
work elsewhere?**

Mistral-7B / C4 — same shape, one real wrinkle



Same free-region-then-monotonic-cost shape seen at every smaller scale.

One non-monotonicity: $\lambda=0.3 \rightarrow 0.4$ prunes *more* for *less* cost — not a real reversal.

$\lambda=0.4$'s convergence trigger simply fired much later (4200 vs. 1150 steps), so it completed far more of its trajectory before the check stopped it. ▶ Same mechanism as the capacity-axis finding from

GPT-2/OPT-125M (F3) — window validity depends on how much of the run has actually finished, now showing up along λ instead of capacity.

- Dense baseline ppl: **4.740** (WikiText-2) vs. **8.325** (C4) — Mistral is already well-calibrated to clean Wikipedia-style text.
- Consequence: much less redundancy to find on WikiText-2 specifically — the whole sparsity/cost curve sits closer to the axis than C4's.

► Exactly the concern flagged *before* running it: a model already well-matched to a domain has less slack for a pruner to discover in the first place.

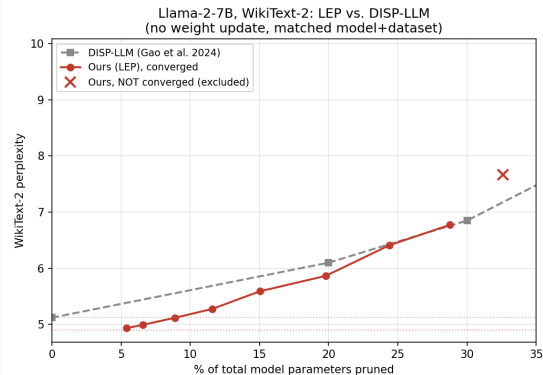
Beating DISP-LLM

Setup: the one directly matched comparison available

- DISP-LLM (Gao 2024) reports LLaMA-2-7B, WikiText-2, **no weight update** — same model, same dataset, same “frozen weights” framing this project already uses.
- Our 9- λ convergence-based sweep, converted to **total-param-pruned %** via Llama-2-7B's 64.24% MLP parameter share, plotted directly against DISP-LLM's own published Table 1 numbers (linearly interpolated between their 5 reported points).

► No re-running DISP-LLM's code, no proxy dataset — their own numbers, our own numbers, same axis.

8 of 8 converged points beat the interpolated DISP-LLM curve

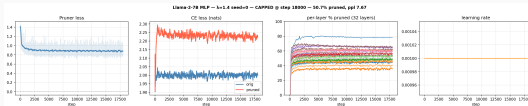


Dense ppl: ours **4.903** vs. DISP-LLM's **5.12**.

- -0.45 ppl at the low end (5.4–6.6% total pruned)
- narrowing to a near-tie ($+0.016$ ppl) at 28.77%

► Every converged point beats it. The margin shrinks as sparsity grows, but never flips negative for any point that actually converged.

The 9th point — diagnosed, not hidden



$\lambda=1.4$ (32.6% total pruned) loses to DISP-LLM by +0.494 ppl — but it **never converged**: pre_decay for the entire 18,000-step cap.

pct_pruned still swings an 8.2pp band in the *final* 20 checkpoints; individual layers still move ~30–48pp step-to-step at step 18,000. A sustained oscillation, not a slow approach to a plateau. ► Excluded from the headline claim on that basis — not a cherry-picked drop.

Why it oscillates, and why “bigger pruner” isn’t the fix

- At $\lambda=1.4$ the sparsity term ($\approx 0.66\text{--}0.8$) is now comparable in size to the CE-cost term ($\approx 0.37\text{--}0.46$) — and LR is held at a constant 10^{-3} regardless of λ , since B9’s decay never gets to intervene without a raw signal firing first.
- Directly tested “increase pruner capacity” against this project’s own prior evidence (an 8-point matched grid at half/base/ $2.26\times$ capacity, GPT-2+OPT-125M) — **no monotonic capacity effect exists there** on %pruned, ppl, or convergence speed.

► Capacity controls the scorer’s expressiveness, not the STE threshold’s oscillation dynamics under a fixed-LR gradient. The indicated fix is LR/ λ -coupling — an optimization-dynamics diagnosis, not a capacity one. Not yet tested at 7B.

**Downstream eval: the ppl win
doesn't fully transfer**

- PIQA / HellaSwag / WinoGrande / ARC-e / ARC-c / BoolQ / OpenBookQA, run on all 8 converged checkpoints + dense.
- Restricted here to DISP-LLM's own 5-task subset (WinoGrande acc; HellaSwag/ARC-e/ARC-c/PIQA acc-norm) for a like-for-like comparison.

► Closes a real gap this project's own venue-analysis flagged: perplexity alone isn't what the comparable literature reports as headline evidence.

Near-parity at light pruning, a growing deficit at heavy pruning

% total params pruned	LEP avg acc	DISP-LLM (interp.)	LEP — DISP-LLM
0% (dense)	68.46	68.99	−0.53
5.39%	66.59	67.03	−0.44
6.57%	66.91	66.61	+0.30
8.89%	66.28	65.76	+0.52
11.58%	64.37	64.79	−0.42
15.05%	62.87	63.53	−0.66
19.79%	60.29	61.81	−1.52
24.38%	59.28	60.14	−0.86
28.77%	55.51	58.55	− 3.04

► Against DISP-LLM's own *actually-measured* 30% point (58.10, a slightly harder ratio than our 28.77%) — still behind by **2.59pp**, direct comparison, no interpolation.

The opposite shape from the perplexity result

Perplexity (earlier): LEP starts ahead, margin *narrows* toward high sparsity.

Downstream accuracy (here): LEP starts at parity, gap *opens up* toward high sparsity.

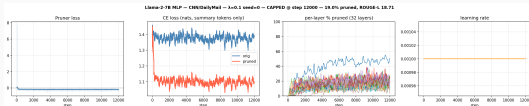
Not yet explained

Candidate, untested: ppl is smooth and measured in-domain (WikiText-2, what the pruner was trained against); the downstream suite is discrete, out-of-domain, and may lean on specific circuits an in-domain-ppl-optimized mask doesn't protect.

- Reported plainly because a reader who only sees the ppl table would draw a stronger conclusion than the data supports.

A first look: pruning for a downstream task

CNN/DailyMail — one point, not a result yet



Target-only CE-delta objective (article masked from the loss, CE on the gold summary only) — same training mechanism as WikiText-2, ported to a real downstream task.

$\lambda=0.1$, single seed, **capped** at 12,000 steps (never converged).

- test ppl: 3.750 $\rightarrow$ 2.881
- ROUGE-L: 13.31 $\rightarrow$ 18.71

► Pruned beats dense on every metric — flagged, not claimed: single non-converged run, no dense-fine-tune control yet, dense baseline itself may just be weak (non-instruction-tuned model, bare completion prompt).

Summary

- Same pruner, zero architecture changes, ported to 7B scale across two model families and three datasets.
- **Convergence is now a checked fact**, not an assumed step count — and the one point that fails the check is diagnosed mechanistically (LR/ λ coupling), not swept under the rug.
- **Beats DISP-LLM's own published numbers at every converged Llama-2-7B/WikiText-2 operating point.**
- **Downstream zero-shot accuracy tells a different story:** near-parity at light pruning, a real and growing deficit at heavy pruning — reported honestly alongside the ppl win, not instead of it.
- First downstream-task pruning run (CNN/DailyMail) shows a striking single-run result — explicitly a case study in progress, gated on a dense-fine-tune control before it means anything.

Thank you.

Findings: `diary/crisp-findings.md` — Paper direction:
`diary/final-paper-direction.md`